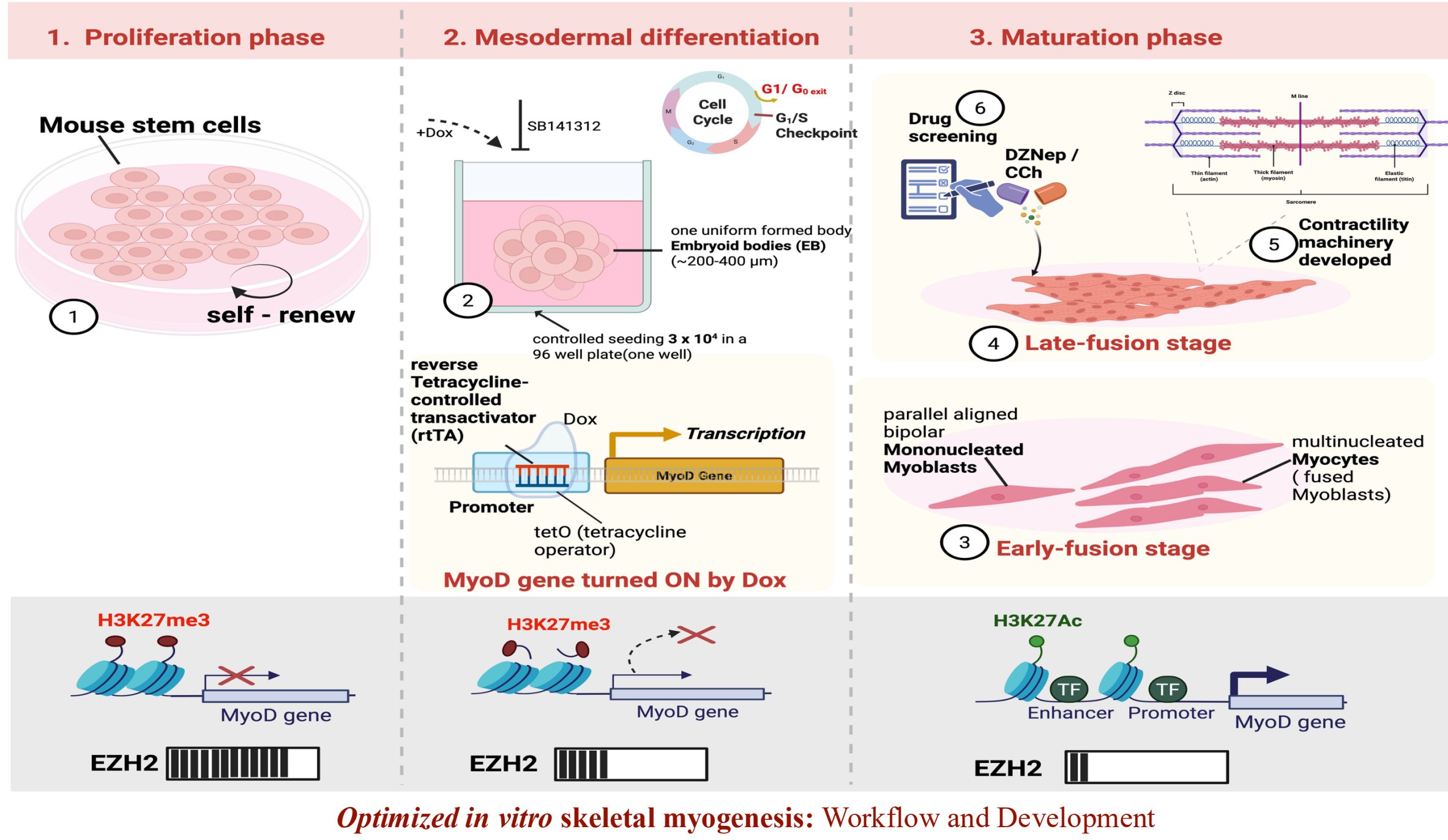


Optimizing in vitro mouse skeletal myogenesis and atrophy mitigation: Pharmacological screening of 3-Deazaneplanocin A (DZNep)

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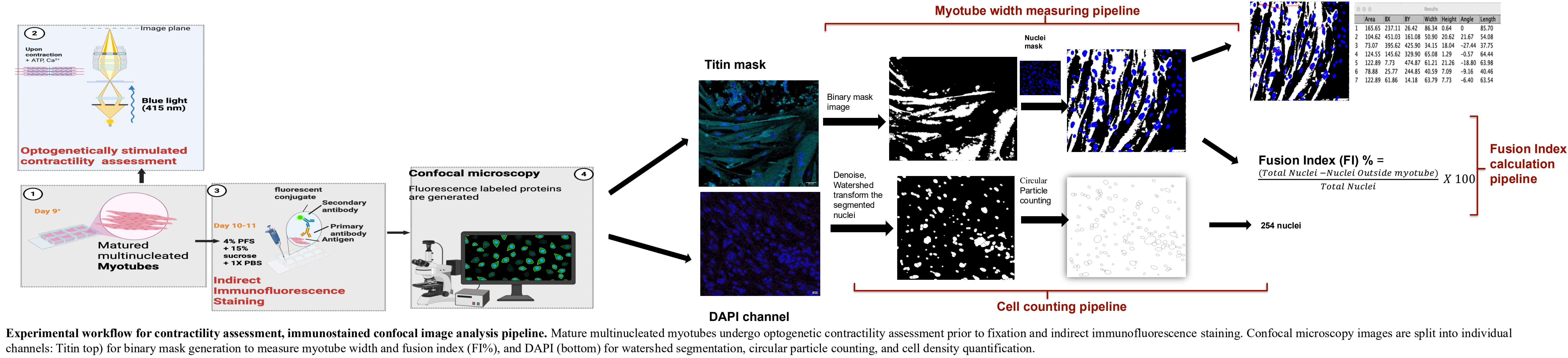
1. Introduction



2. Aims

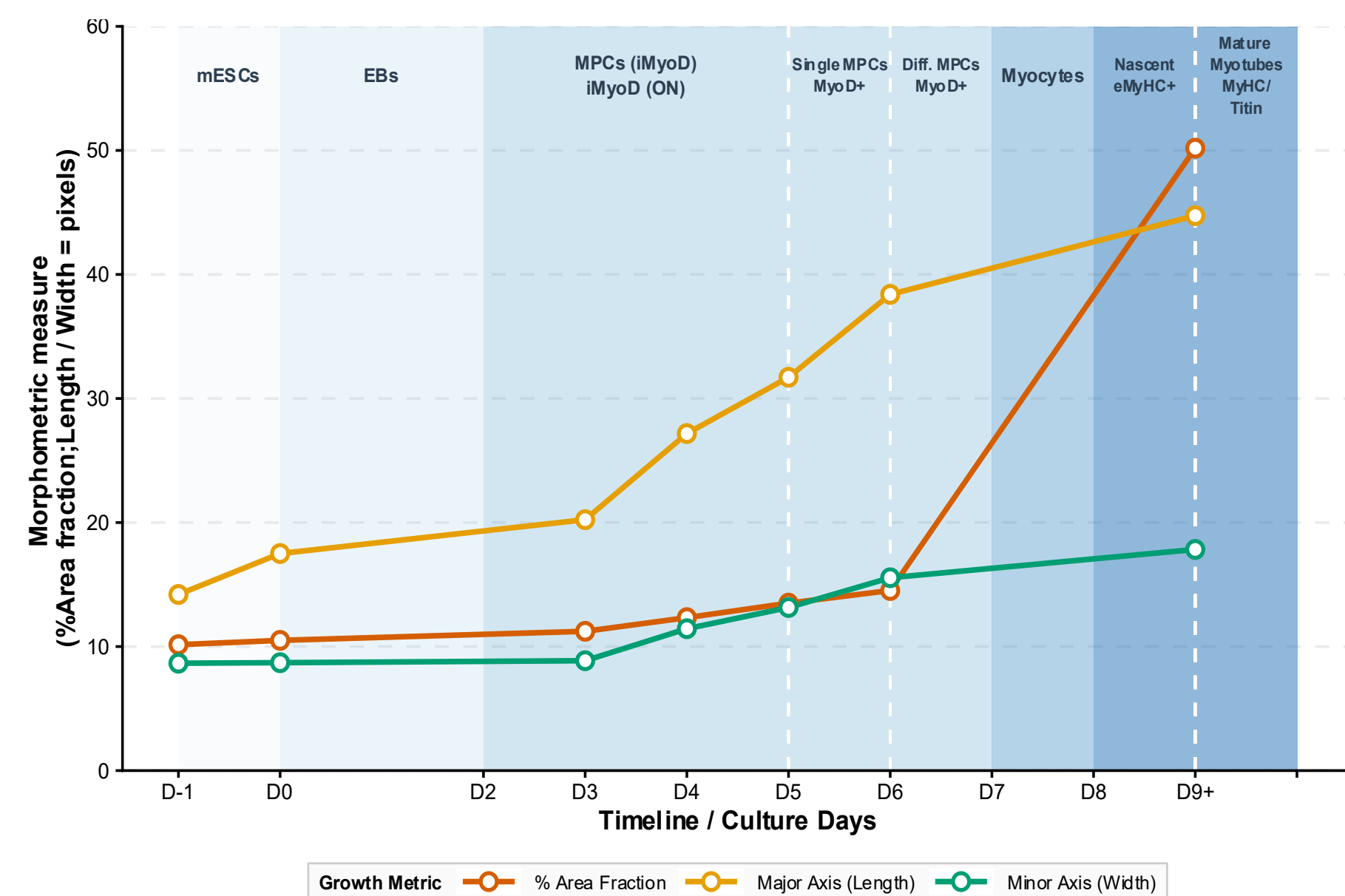
1. Optimize *in vitro* mouse skeletal myogenesis for homogenous cultures.
2. Evaluate the effects of substrates on myotube maturation.
3. Evaluate Glut4 (Medium vs High) expression lines for glucose metabolism studies.
4. Mitigation of Atrophy via Carbachol (CCh) induction.
5. Evaluate DZNep dose-dependent effect on myotube morphology and viability.

3. Methods

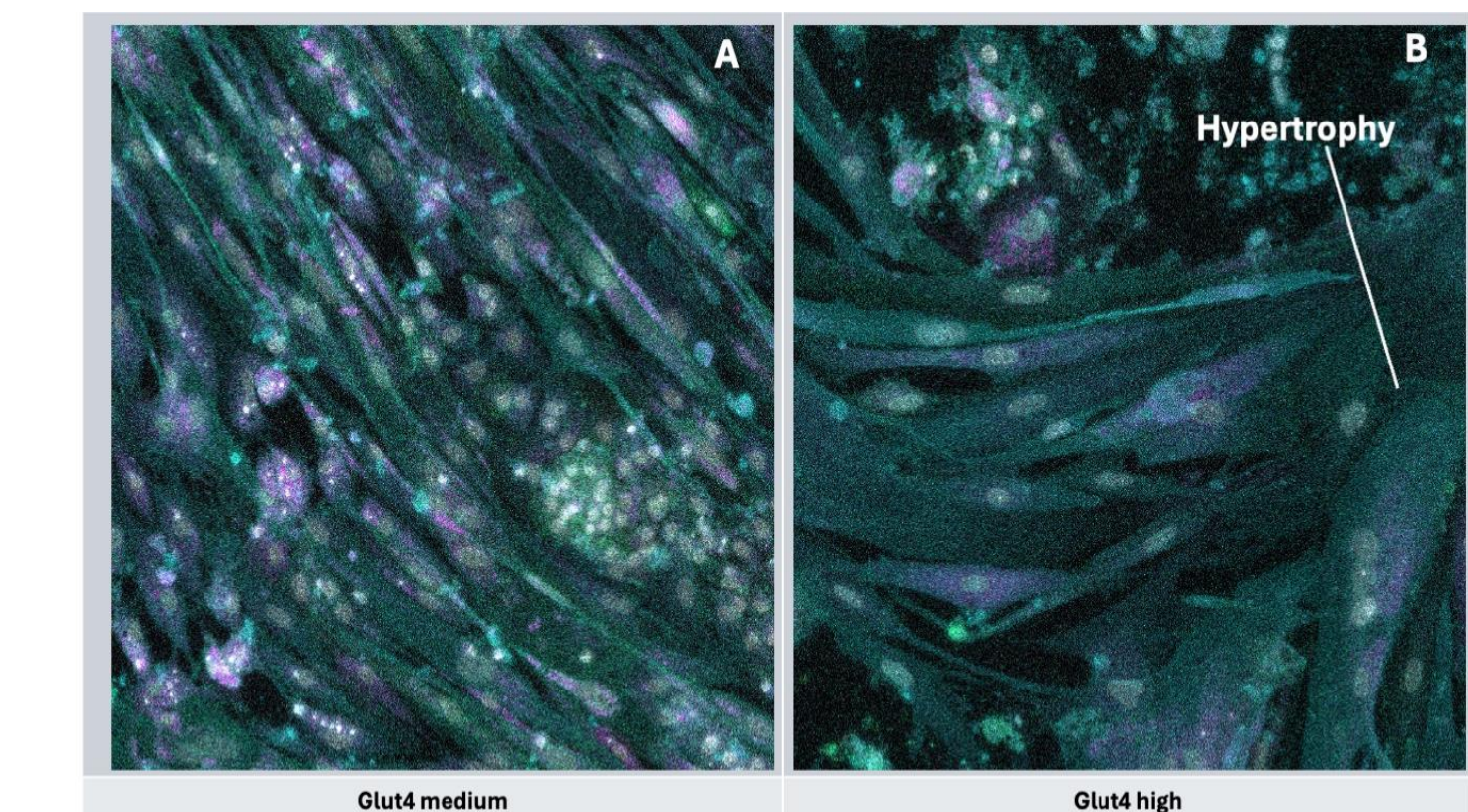


4. Results

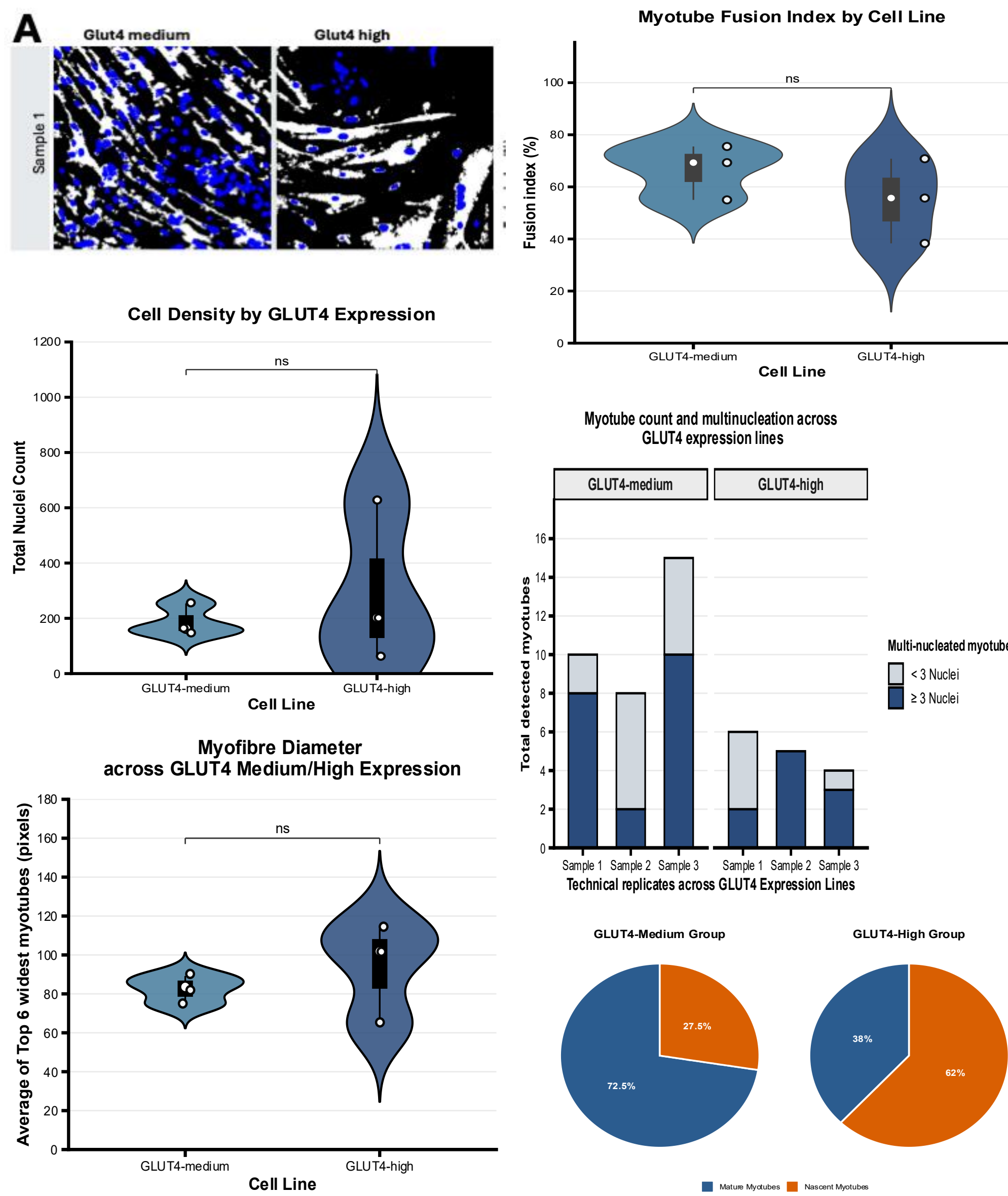
1. Glut4 medium vs Glut4 high overexpression cell lines



Canonical Morphological Trajectory of the cultured *In Vitro* Skeletal Myogenesis indicated that late-stage myogenesis is predominantly characterized by cell fusion. Myotube elongation (Major Axis/Length) progresses linearly throughout the culture timeline (reaching ~45 pixels at D9+) whereas the myotube width (Minor Axis) exhibits controlled, linear enlargement (8 to 18 pixels). Overall, cell coverage (% Area Fraction) undergoes a dramatic non-linear surge following single MPC or myocyte fusion (D6 to D9+).

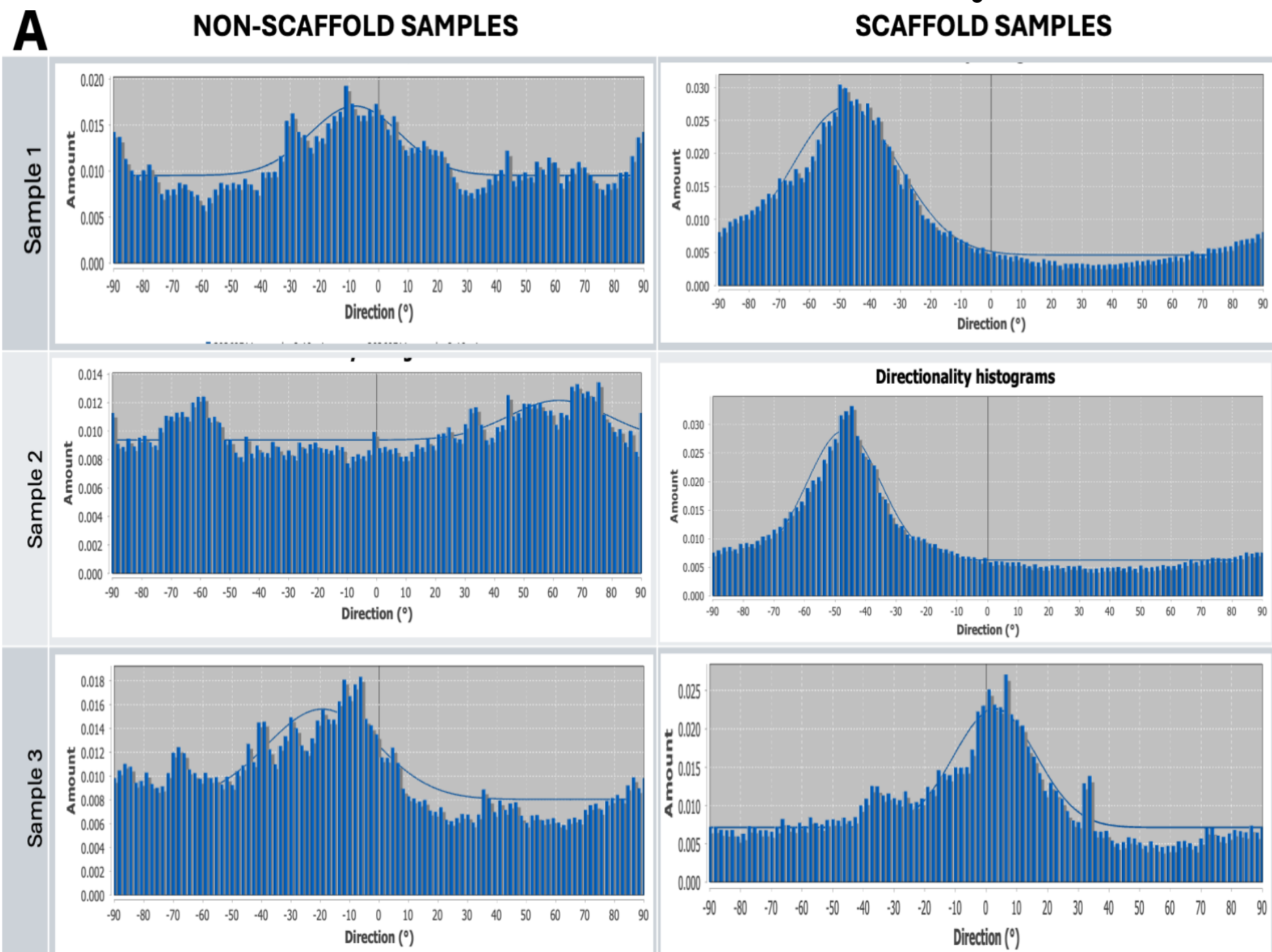


(A) GLUT4-Medium demonstrated uniform, linear alignment of mature myotubes whereas GLUT4 High (B) depict evident cellular hypertrophy and a higher proportion of unfused cells. (Stains: Titin = green, DAPI = white, GLUT4 = magenta, Myosin = cyan)

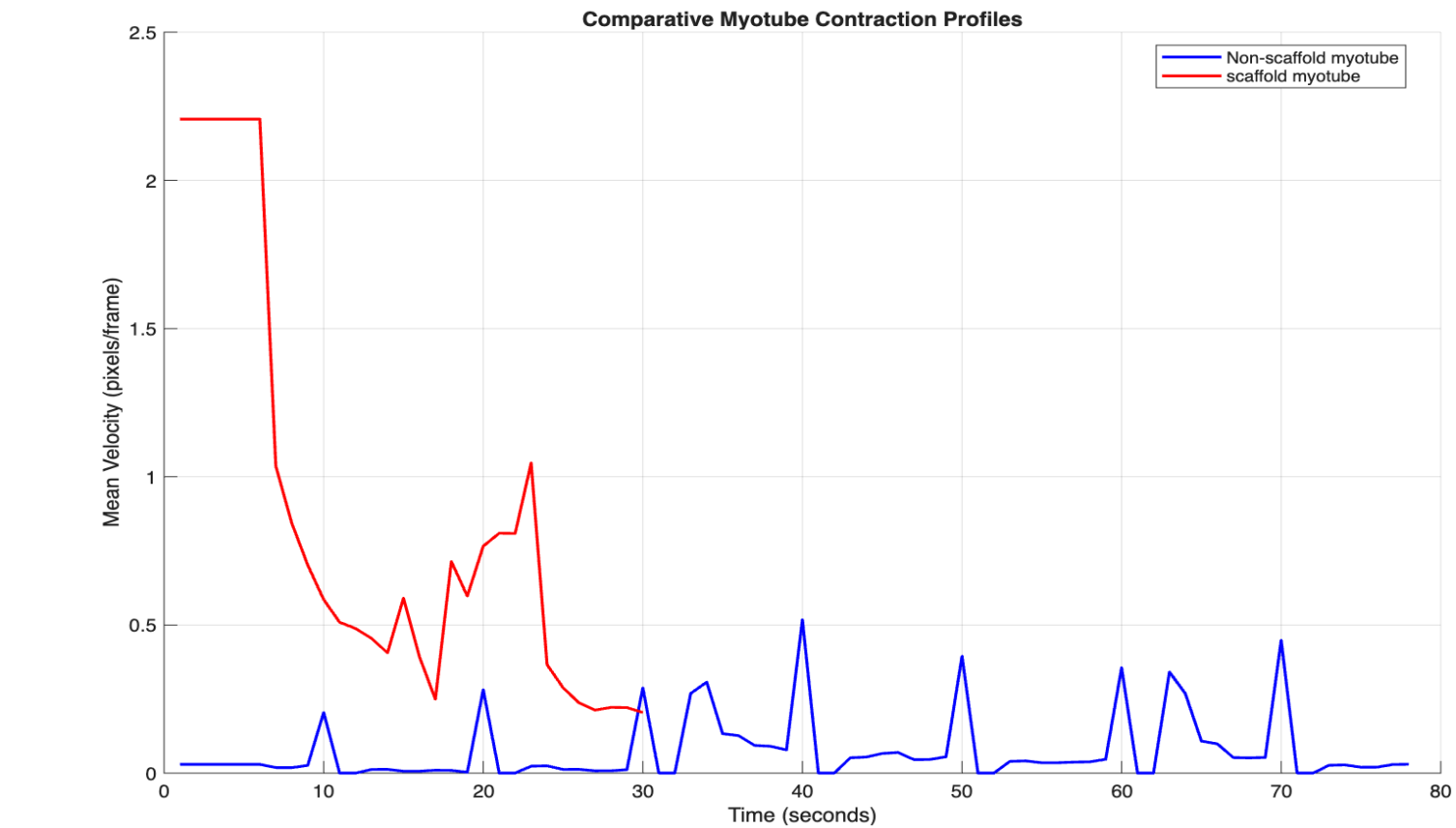


Excessive overexpression of GLUT4 impedes myotube fusion. The GLUT4-Medium exhibits an optimal fused-to-unfused (mononucleated) cell ratio (72.5% mature vs nascent myotubes 27.5%) with uniform myotube alignment along with high cell density whereas GLUT4 high exhibited (38.0% mature vs 62.0% nascent) accompanied with hypertrophy and atypical fusion kinetics along accompanied with reduced cell density masquerading the rate of fusion. SB=50 μ m.

2. Non-Scaffold vs Scaffold cultured myotubes

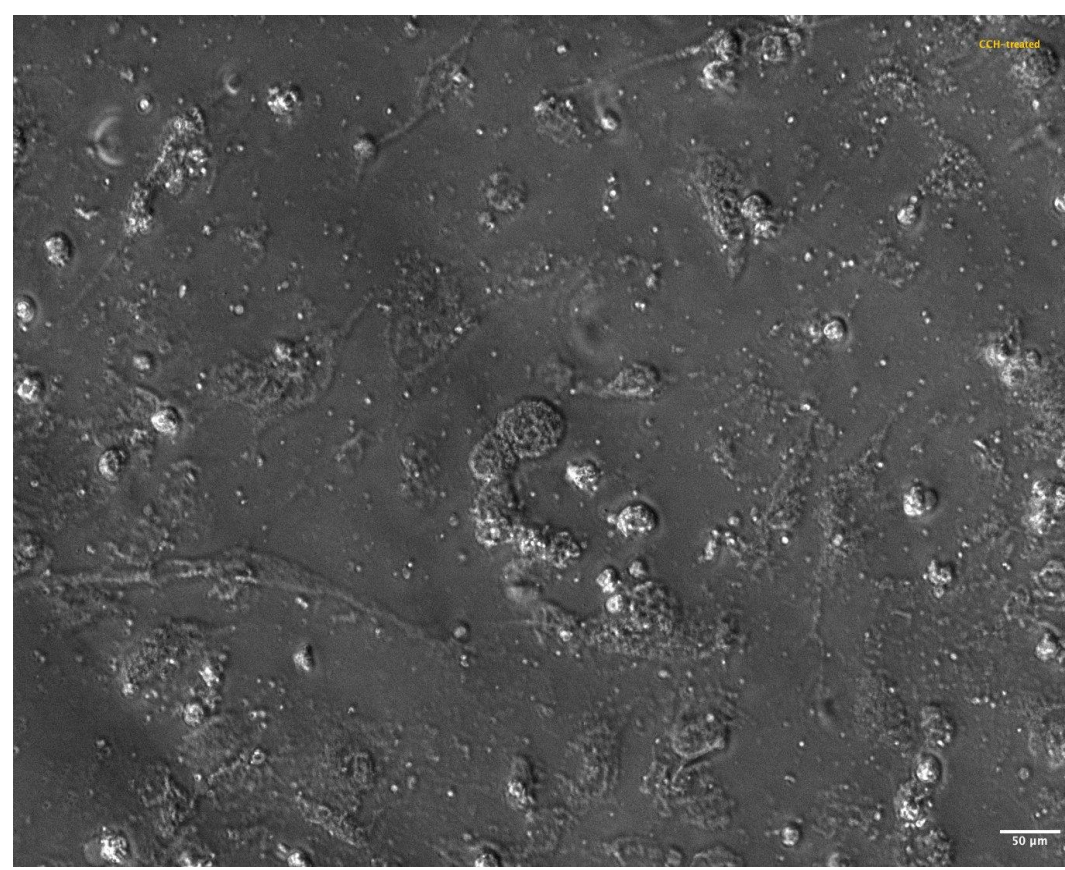


The scaffold substrates successfully imposed unidirectional cellular orientation (anisotropic alignment). In the absence of scaffolds, non-scaffold myotubes exhibited a random, isotropic orientation.



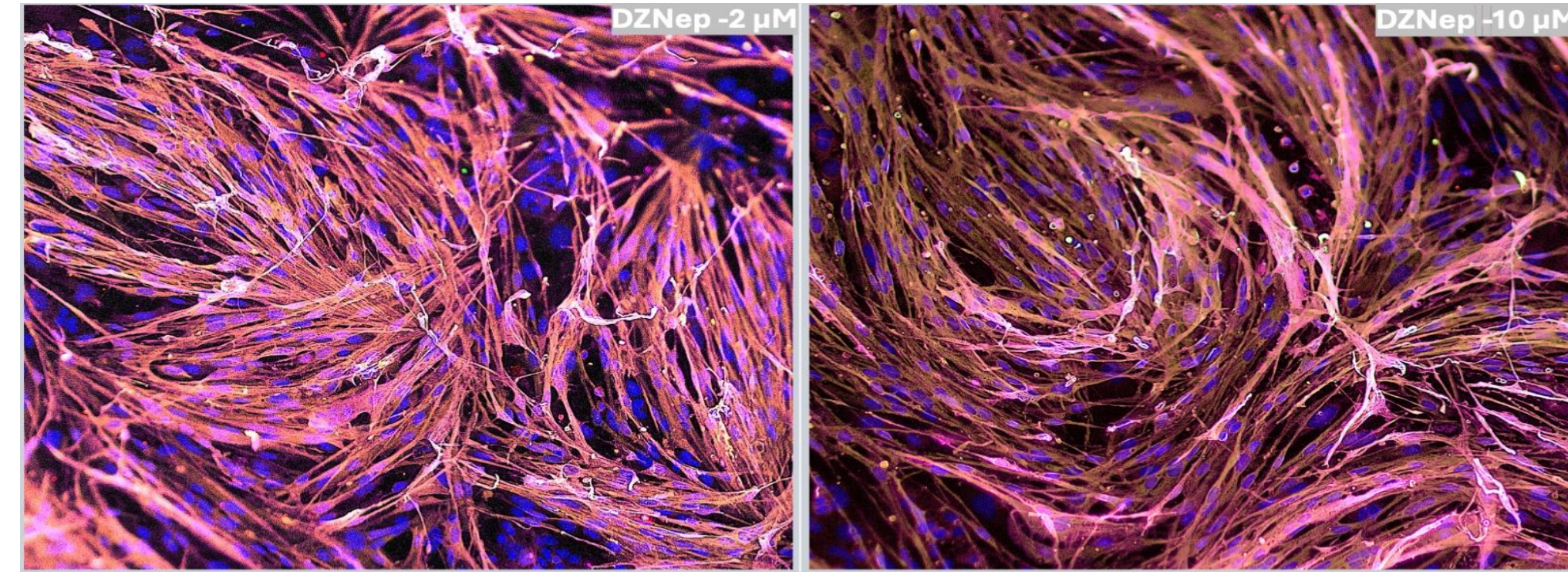
Structural scaffolding is essential mechanical force generation. Non-scaffold myotubes exhibiting low amplitude mean contraction velocity (~0.5 pixels/frame) in comparison to anisotropic scaffold-cultured myotubes (~2.2 pixels/frame), demonstrating that structural scaffolding is essential mechanical force generation.

3. Carbachol induction

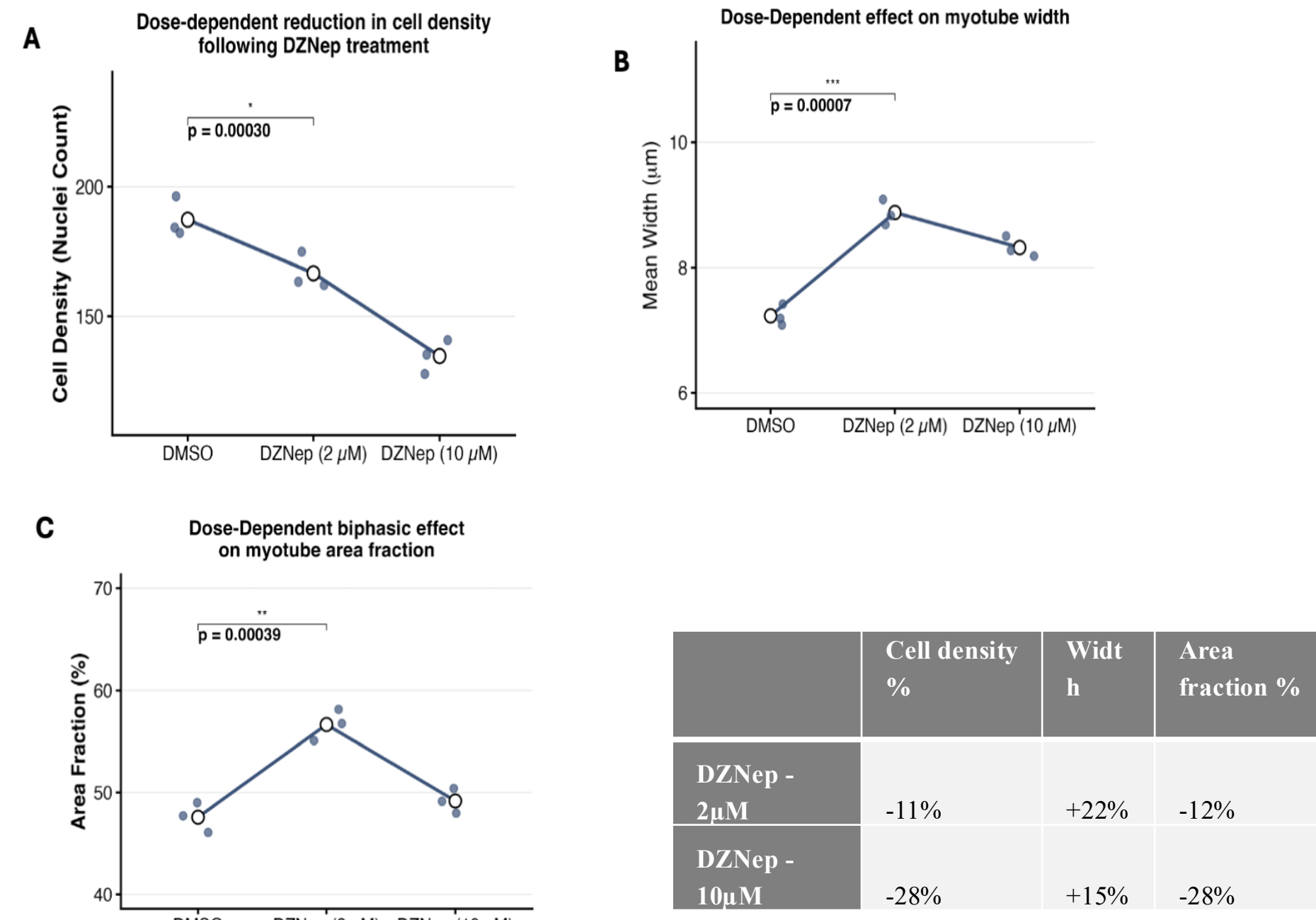


Premature exposure during early-stage induction to CCh causes cholinergic cytotoxicity and receptor endocytosis (vacuolization). Thus, exposure must be restricted to late-stage maturation (Days 9–11).

4. Dose-dependent effect on maturation of myotubes



Immunofluorescence staining confirms DZNep-Induced Morphological and Structural Alterations. At 2 μ M, DZNep, myotubes exhibit dense, elongated myofibrillar networks with robust nuclear alignment. In contrast, treatment with 10 μ M DZNep leads to noticeable myofibrillar fragmentation (blunt boundaries), thinning of muscle fibers, and a marked reduction in nuclear density (blue, DAPI).



(A) Nuclear count: Dose-dependent monotonic decrease in cell density as DZNep concentration increases, marked decrease at 2 μ M (compared to DMSO: $p = 0.00030$) and further declining at 10 μ M. (B) Mean Width: Displays a non-linear response where DZNep-2 μ M significantly increases myotube width compared to DMSO control ($p = 0.00007$), but DZNep-10 μ M leads to a slight decline relative to the 2 μ M peak. (C) Area Fraction: Demonstrates a clear biphasic effect where total coverage increases from DMSO (~48%) to 2 μ M (~57%, $p = 0.00039$), but drops back down toward control levels at 10 μ M (~49%).

5. Conclusion

1. Optimization *in vitro* skeletal myogenesis via Controlled and uniform seeding density (3×10^4 cells/well) produced intermediate EB of size (200–400 μ m) overcoming 2D substrate barriers yielding highly synchronized, multinucleated myotube.

2. GLUT4-Medium expression is the ideal candidate for glucose metabolism studies with robust myoblast fusion, high cell density and high fusion rate (≥ 3 nuclei; 72.5%) without overexpression artifacts. High GLUT4 expression impedes myogenic fusion. Conversely, GLUT4-High depicted Compensatory Cellular Hypertrophy (CCH), leaving 62% of the myotubes in a nascent state.

3. Anisotropic nanofibrous scaffolds impose unidirectional mechanical alignment required to produce synchronized and high-velocity contraction (2.2 vs 0.5 pixels/frame in 2D plastic).
4. Premature carbachol exposure (early-stage induction) induce cholinergic cytotoxicity and receptor endocytosis (vacuolization), indicating that treatment must be restricted to late-stage maturation (Days 9–11).

5. DZNep induced dose-dependent nuclear loss accompanied by vacuolar degeneration, not true myofibrillar hypertrophy. (DZNep-2 μ M nuclear loss: -11%; width: +22%; Area fraction: -12% at 2 μ M whereas DZNep-10 μ M nuclear loss: -28%; width: +15%; Area fraction: -28%; at 10 μ M)

6. Limitations/ Future Research

1. 2D monocultures restrict long-term studies due to cell detachment; future research must integrate cost-effective 3D scaffolds
2. Brightfield imaging cannot distinguish genuine hypertrophy from pathological swelling (DZNep); experiments must be repeated using myosin-targeted immunostaining
3. Early-stage carbachol exposure induces cytotoxicity, necessitating drug treatment at late-stage maturation